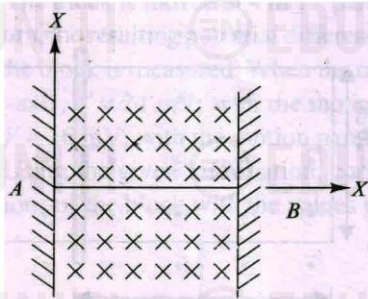


### Paragraph Q11 to Q13

A standing wave  $y = 2A \sin kx \cos \omega t$  is set up in the wire  $AB$  fixed at both ends by two vertical walls. The region between the walls contains a constant magnetic field  $B$ . Now answer the following questions.



11. The wire is found to vibrate in the third harmonic. The maximum emf induced is
 

(a)  $\frac{4(AB)\omega}{k}$

(b)  $\frac{3(AB)\omega}{k}$

(c)  $\frac{2(AB)\omega}{k}$

(d)  $\frac{(AB)\omega}{k}$
12. In the above question, the time when the emf becomes maximum for the first time is
 

(a)  $\frac{2\pi}{\omega}$

(b)  $\frac{\pi}{\omega}$

(c)  $\frac{\pi}{2\omega}$

(d)  $\frac{\pi}{4\omega}$
13. In which of the following modes the emf induced in  $AB$  is always zero
 

(a) fundamental mode

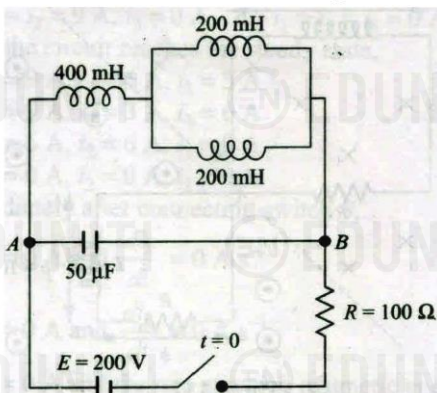
(b) second harmonic

(c) second overtone

(d) fourth overtone

### Paragraph Q14 to Q15

In the circuit shown in Fig., switch  $S_1$  was closed for a long time. At time  $t = 0$  the switch is opened.



14. Find the maximum potential difference across the plates of the capacitor after the switch is opened.
 

(a) 100 V

(b) 200 V

(c) 300 V

(d) 400 V

15. Find the angular frequency of oscillation of the charge on the capacitor
 

(a) 100 rad s<sup>-1</sup>

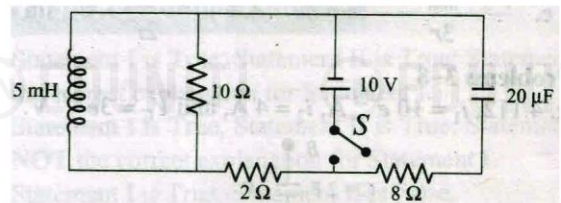
(b) 200 rad s<sup>-1</sup>

(c) 300 rad s<sup>-1</sup>

(d) 400 rad s<sup>-1</sup>

### Paragraph Q16 to Q18

In the given circuit at  $t = 0$ , switch  $S$  is closed.



16. The current through the  $10 \Omega$  resistor at any instant  $t$  ( $0 < t < \infty$ ) will be
 

(a)  $\frac{1}{6}e^{-(1000/3)t}$

(b)  $\frac{5}{6}e^{-(1000/3)t}$

(c)  $\frac{1}{6}e^{(1000/3)t}$

(d)  $\frac{6}{5}e^{(1000/3)t}$
17. The energy stored in the inductor at any instant  $t$  ( $0 < t < \infty$ ) will be
 

(a)  $\frac{1}{2}[5 - 5e^{-(1000/3)t}]^2$  mJ

(b)  $\frac{125}{2}[1 - e^{-(1000/3)t}]^2$  mJ

(c)  $\frac{25}{2}[1 - e^{-(1000/3)t}]^2$  mJ

(d)  $\frac{5}{2}[1 - e^{-(1000/3)t}]^2$  mJ
18. The energy stored in the capacitor and inductor, respectively, at  $t \rightarrow \infty$  will be
 

(a) 1 mJ and 62.5 mJ

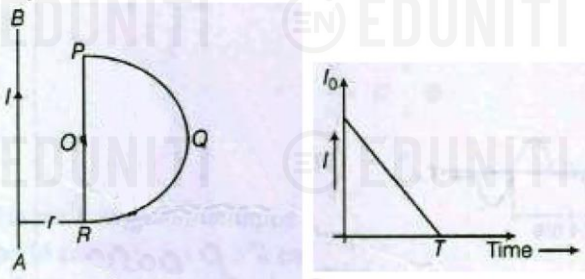
(b) 62.5 mJ and 1 mJ

(c) 2 mJ and 62.5 mJ

(d) 1 mJ and 60 mJ

## Paragraph Q19 to Q21

A semicircular wire with  $PQROP$  has a resistance  $R$ . Its radius  $r$  ( $OP = OQ = OR = r$ ) with  $POR$  as the diameter at a distance  $r$  from a long parallel wire  $AB$  carrying a time current  $I$ , as shown in the figure. The current in wire  $AB$  decreases linearly from an initial value  $I_0$  to zero in time  $T$  ( $Q$  is charge flown in wire in time  $T$ )



19. Find magnetic flux through  $PQROP$ , when current  $I$  in wire  $AB$  is

- (a)  $\frac{\mu_0 I r}{2\pi}(\pi - 2)$  (b)  $\frac{2\mu_0 I r}{\pi}(\pi - 2)$   
 (c)  $\frac{\mu_0 I r}{\pi}(\pi - 2)$  (d) None of these

20. Find induced current in  $PQROP$  as a function of time.

- (a)  $\frac{\mu_0 r(\pi - 2)Q}{\pi R T^2}$  (b)  $\frac{\mu_0 r(\pi - 2)Q}{2\pi T^3}$   
 (c)  $\frac{2\mu_0 r(\pi - 2)^2 Q}{\pi R T^2}$  (d) None of these

21. Find heat generated in  $PQROP$  in time  $T$ .

- (a)  $\frac{\mu_0 r^2(\pi - 2)^2 Q^2}{\pi R T^3}$  (b)  $\frac{\mu_0^2 r^2(\pi - 2)^2 Q^2}{\pi^2 R T^3}$   
 (c)  $\frac{\mu_0 r^2(\pi - 2)^2 Q^2}{\pi^2 R T^2}$  (d) None of these

## Paragraph Q22 to Q24

A solenoid of resistance  $R$  and inductance  $L$  has a piece of soft iron inside it. A battery of emf  $E$  and of negligible internal resistance is connected across the solenoid as shown in the figure. At an instant, the piece of soft iron is pulled out suddenly so that inductance of the solenoid decreased to  $\eta L$  with battery remain connected.

22. Find the work done to pull out the soft iron piece.

- (a)  $\frac{\eta L E^2}{2R^2}$  (b)  $\frac{(1 - \eta) L E^2}{2R^2}$   
 (c)  $\frac{(1 - \eta) L E^2}{\eta R^2}$  (d)  $\frac{(1 - \eta) L E^2}{2\eta R^2}$

23. Assuming  $t = 0$  is the instant when iron piece has been pulled out, the current as a function of time is

- (a)  $i = \frac{E}{R} \left( 1 - \left( 1 - \frac{1}{\eta} \right) e^{-t/\tau} \right)$  (b)  $i = \frac{E}{R} \left( 1 + \left( 1 + \frac{1}{\eta} \right) e^{-t/\tau} \right)$   
 (c)  $i = \frac{E}{R} \left( 1 - \left( 1 + \frac{1}{\eta} \right) e^{-t/\tau} \right)$   
 (d)  $i = \frac{E}{R} \left( 1 + \left( 1 - \frac{1}{\eta} \right) e^{-t/\tau} \right)$  (where,  $\tau = \frac{\eta L}{R}$ )

24. Find power supplied by the battery as a function of time.

- (a)  $P = \frac{E^2}{R} \left( 1 - \left( 1 - \frac{1}{\eta} \right) e^{-t/\tau} \right)$   
 (b)  $P = \frac{E^2}{R} \left( 1 + \left( 1 + \frac{1}{\eta} \right) e^{-t/\tau} \right)$   
 (c)  $P = \frac{E^2}{R} \left( 1 - \left( 1 + \frac{1}{\eta} \right) e^{-t/\tau} \right)$   
 (d)  $P = \frac{E^2}{R} \left( 1 + \left( 1 - \frac{1}{\eta} \right) e^{-t/\tau} \right)$

Answer Key

- 11. a
- 12. c
- 13. b
- 14. b
- 15. b
- 16. b
- 17. b
- 18. a
- 19. a
- 20. a
- 21. b
- 22. d
- 23. a
- 24. a